1 Basic For Loops

Problem 1.1. Express the asymptotic running time of the following functions as summations. **You do not need to analyze the sums, just write down summations.**

(a)

```
function Func(A[],n)
    x = 0
    for i = 1 to n^2 do
        for j = i to n^2 do
            x = x + A[⌊i/j⌋]
        end for
    end for
    return x
end function
```

$$\sum_{i=1}^{n^2} \sum_{j=i}^{n^2} c$$

(b)

```
function Func(A[],n)
    x = 0
    for i = 2n + 3 to 5n^2 + 3n do
        for j = 3n to 6i^2 + 3i + 12 do
            for k = ij to n^10 do
                x = x + A[(ij + k) mod n]
            end for
        end for
    end for
    return x
end function
```

$$\sum_{i=2n+3}^{5n^2+3n} \sum_{j=3n}^{6i^2+3i+12} \sum_{k=ij}^{n^{10}} c$$

(c)

```
function Func(A[],L,U)
    x = 0
    n = U − L + 1
    for i = L to U do
        for j = i to U do
            x = x + A[i] − A[j]
        end for
    end for
    return x
end function
```

$$\sum_{i=L}^{U} \sum_{j=i}^{U} c$$

Problem 1.2. Determine the asymptotic complexity of the given summations.

(a)

$$R(n) = \sum_{k=0}^{3n-\frac{1}{2}} c\log_4(3n-k)$$

UB: $R(n) \leq \sum_{k=0}^{3n-1/2} c\log_2(3n)$ [Replace w/ max]

$\leq (3n - \frac{1}{2} + 1)\, c\log_2(n)$ [Simplify]

LB: $R(n) \geq \sum_{k=0}^{n} c\log_2(3n-k)$ [shrink range]

$\geq \sum_{k=0}^{n} c\log_2(3n-n)$ [replace w/ min]

$\geq (n+1)\, c\log_2(2n)$ [simplify]

Thus, $R(n) \in \Theta(n\log(n))$

(b)

$$S(n) = \sum_{i=1}^{n} \left(\frac{n}{2} - i\right)^2$$

$$S(n) = \underbrace{\sum_{i=0}^{n/2} \left(\frac{n}{2} - i\right)^2}_{\left(\frac{n}{2}\right)^2 + \left(\frac{n}{2}-1\right)^2 + \dots + 0^2} + \underbrace{\sum_{i=\frac{n}{2}+1}^{n} \left(\frac{n}{2} - i\right)^2}_{(-1)^2 + (-2)^2 + \dots + \left(\frac{-n}{2}\right)^2}$$

the same! both $= \sum_{i=1}^{n/2} i^2$

$$= 2 \sum_{i=0}^{n/2} i^2$$

$$= 2 \cdot \frac{\frac{n}{2} \cdot \left(\frac{n}{2}+1\right)\left(2 \cdot \frac{n}{2}+1\right)}{6}$$

Thus, $S(n) \in \Theta(n^3)$.

OR

$$\sum_{i=0}^{n} \left(\frac{n}{2} - i\right)^2 = \sum_{i=0}^{n} \left(\frac{n^2}{4} - ni + i^2\right)$$

$$= \sum_{i=0}^{n} \frac{n^2}{4} - \sum_{i=0}^{n} ni + \sum_{i=0}^{n} i^2$$

$$= n \cdot \frac{n^2}{4} - n \cdot \frac{n(n+1)}{2} + \frac{n(n+1)(2n+1)}{6}$$

$$\approx \frac{n^3}{4} - \frac{n^3}{2} + \frac{n^3}{3} = \frac{n^3}{12}$$

OR

$$S(n) \le \sum_{i=0}^{n} \left(\frac{n}{2}\right)^2$$

$$\le (n+1) \cdot \frac{n^2}{4}$$

$$S(n) \ge \sum_{i=0}^{n/4} \left(\frac{n}{2} - i\right)^2$$

$$\ge \sum_{i=0}^{n/4} \left(\frac{n}{2} - \frac{n}{4}\right)^2$$

$$\ge \left(\frac{n}{4} + 1\right) \cdot \left(\frac{n}{4}\right)^2$$

Problem 1.3. Determine the asymptotic complexity of the following function.

```
function Func(A[],L,U)
    x = 0
    n = U - L + 1
    for i = L to U do
        for j = i to U do
            x = x + A[i] - A[j]
        end for
    end for
    return x
end function
```

Hint: The U and L variables suck. Try to find a way to change the problem to only have n.

Hint: Writing out the terms of the sum may help. For example with a *different* sum,

$$\sum_{i=0}^{n}(n-i)^2 = n^2 + (n-1)^2 + (n-2)^2 + \cdots + (n-(n-2))^2 + (n-(n-1))^2 + (n-n)^2$$
$$= 0^2 + 1^2 + 2^2 + \cdots + (n-2)^2 + (n-1)^2 + n^2 = \sum_{i=1}^{n} i^2$$

might make the sum easier.

$$T(n) = \sum_{i=L}^{U}\sum_{j=i}^{U} c = c\sum_{i=L}^{U}(U-i+1)$$

$$= c\Big[\overbrace{(U-L+1)}^{n} + \overbrace{(U-(L+1)+1)}^{n-1} + \overbrace{(U-(L+2)+1)}^{n-2}$$
$$+ \cdots + \underbrace{(U-(U-2)+1)}_{3} + \underbrace{(U-(U-1)+1)}_{2} + \underbrace{(U-U+1)}_{1}\Big]$$

$$= c[n + (n-1) + \cdots + 2 + 1]$$ // simplify using $n = U - L + 1$

$$= c[1 + 2 + \cdots + (n-1) + n]$$ // reverse the sum

$$= c\sum_{i=1}^{n} i = c\,\frac{n(n+1)}{2}$$

Thus, $T(n) \in \Theta(n^2)$.

2 Basic While Loops

Problem 2.1. Express the running time of the following functions as summations.
You do not need to analyze the sums, just write down summations.

(a)

```
function Func(A[],n)
    x = 0
    for i = 1 to n^2 do
        j = i^2
        while j ≥ i do
            x = x + i − j
            j = j − 10
        end while
    end for
end function
```

Iteration	j
0	i^2
1	i^2-10
2	i^2-20
$\vdots$	$\vdots$
k	i^2-10k

Stops when

$i^2-10k' = i$

$k' = \frac{i^2-i}{10}$

$$\sum_{i=1}^{n^2} c\left(\frac{i^2-i}{10}\right)$$

(b)

```
function Func(A[],n)
    x = 0
    for i = n to n^2 do
        j = i^3
        while j ≥ i do
            x = x + i − j
            j = ⌊j/2⌋
        end while
    end for
end function
```

Iteration	j
0	i^3
1	$i^3/2$
2	$i^3/4$
$\vdots$	$\vdots$
k	$i^3/2^k$

Stops when

$\frac{i^3}{2^{k'}} = i$

$k' = \log_2(i^2)$

$$\sum_{i=n}^{n^2} c\log_2(i^2)$$

(c)

```
function Func(A[],n)
    x = 0
    for i = 1 to n do
        j = 1
        while j ≤ √i do
            x = x + i − j
            j = 5j
        end while
    end for
end function
```

Iteration	j
0	1
1	5
2	25
⋮	
k	5^k

Stops when

$5^{k'} = \sqrt{i}$

$k' = \log_5(\sqrt{i})$

$$\sum_{i=1}^{n} c \log_5(\sqrt{i})$$

(d)

```
function Func(A[],n)
    x = 0
    for i = 1 to n^3 do
        j = i
        while j ≤ 3i^2 do
            x = x + i − j
            j = j + 2n
        end while
    end for
end function
```

Iteration	j
0	i
1	$i+2n$
2	$i+4n$
⋮	⋮
k	$i+2kn$

Stops when

$i + 2k'n = 3i^2$

$k' = \frac{3i^2 - i}{2n}$

$$\sum_{i=1}^{n^3} c\left(\frac{3i^2 - i}{2n}\right)$$

Problem 2.2. Determine the asymptotic complexity of the given summations.

(a)

$$T(n) = \sum_{j=1}^{n^4} \log_3(2n^4 - j)$$

UB:

$$T(n) \leq \sum_{j=1}^{n^4} \log_3(2n^4)$$

$$\leq n^4 \cdot \log_3(2n^4)$$

LB:

$$T(n) \geq \sum_{j=1}^{n^4} \log_3(2n^4 - n^4)$$

$$\geq n^4 \cdot \log_3(n^4).$$

Thus, $T(n) \in \Theta(n^4 \log(n))$.

(b)

$$T(n) = \sum_{i=1}^{\log_2(n)} c\log_2(n)\sqrt{i^2+2}$$

UB:

$$T(n) \le \sum_{i=1}^{\log_2(n)} c\log_2(n)\sqrt{(\log_2(n))^2+2}$$

$$T(n) \le \log_2(n) \cdot c\log_2(n)\sqrt{(\log_2(n))^2+2}$$

LB:

$$T(n) \ge \sum_{i=\frac{\log_2(n)}{2}}^{\log_2(n)} c\log_2(n)\cdot\cancel{\sqrt{i^2}}\; i$$

$$\ge \sum_{i=\frac{\log_2(n)}{2}}^{\log_2(n)} c\log_2(n)\cdot\frac{\log_2(n)}{2}$$

$$\ge \left(\log_2(n) - \frac{\log_2(n)}{2} + 1\right)\cdot c\log_2(n)\cdot\frac{\log_2(n)}{2}$$

Thus, $T(n) \in \Theta(\log(n)^3)$

NOTE: $\log(n)^3 \neq \log(n^3)$.

Inputs to functions MATTER!

Problem 2.3. Determine the asymptotic complexity of the following function.

```
function Func(A[],n)
    x = 0
    for i = 1 to n^3 do
        x = x + Foo(A[], i^3)
    end for
end function
```

$\Theta(\sqrt{i^3})$

$$T(n) = \sum_{i=1}^{n^3} c\sqrt{i^3}$$

where you know that $\texttt{Foo}(A[], n)$ takes $\Theta(\sqrt{n})$ time.

Hint: This problem isn't *that* hard but its also not *super* easy.

UB:

$$T(n) \le \sum_{i=1}^{n^3} c\sqrt{n^9} = n^3 \cdot cn^{9/2}$$

LB:

$$T(n) \ge \sum_{i=\frac{n^3}{2}}^{n^3} c\sqrt{i^3} \ge \sum_{i=\frac{n^3}{2}}^{n^3} c\sqrt{\left(\frac{n^3}{2}\right)^3} \ge \frac{n^3}{2} \cdot c\sqrt{\frac{n^9}{2^3}}$$

Thus, $T(n) \in \Theta(n^{7.5})$.

3 Advanced While Loops

Problem 3.1. Express the running time of the following functions as a single summation.
YOU DO NOT NEED TO ANALYZE THE SUM.

(a)

```
function Func(A[],n)
    x = 0
    i = 1
    while i ≤ ⌊√n⌋ do
        for j = 1 to i^2 do
            x = x + i − j
        end for
        i = i + 10
    end while
end function
```

ci^2

Iteration	i
0	1
1	11
2	21
⋮	⋮
k	$1 + 10k$

Stops when

$1 + 10k' = \sqrt{n}$

$k' = \frac{\sqrt{n} - 1}{10}$

$$T(n) = \sum_{\substack{\text{Values} \\ \text{of } i}} ci^2 = \sum_{k=0}^{k'} c(1 + 10k)^2$$

(b)

```
function Func(A[],n)
    x = 0
    i = n^2
    while i ≥ n do
        for j = 1 to i^3 do
            x = x + i − j
        end for
        i = ⌊i/6⌋
    end while
end function
```

ci^3

Iteration	i
0	n^2
1	$n^2/6$
2	$n^2/36$
⋮	⋮
k	$n^2/6^k$

Stops when

$\frac{n^2}{6^{k'}} = n$

$k' = \log_6(n)$

$$T(n) = \sum_{\substack{\text{Values} \\ \text{of } i}} ci^3 = \sum_{k=0}^{k'} c(n^2/6^k)^3$$

(c)

```
function Func(A[],n)
    x = 0
    i = 1
    while i ≤ n^3 do
        j = i
        while j ≤ i^2 do
            x = x + i − j
            j = j + i
        end while
        i = 3i
    end while
end function
```

$c(i-1)$

Iteration	i
0	1
1	3
2	9
⋮	⋮
k	3^k

Stops when

$3^{k'} = n^3$

$k' = \log_3(n^3)$

Iteration	j
0	i
1	$2i$
2	$3i$
⋮	⋮
ℓ	$(\ell+1)i$

Stops when

$(\ell'+1)i = i^2$

$\ell' = i-1$

$$T(n) = \sum_{\substack{\text{Values}\\ \text{of } i}} c(i-1) = \sum_{k=0}^{k'} c(3^k - 1)$$

Problem 3.2. Determine the asymptotic complexity of the given summations.

(a)

$$T(n) = \sum_{k=0}^{\log_2(n)} cn \cdot 4^{2k}$$

$$T(n) = cn \sum_{K=0}^{\log_2(n)} 16^K$$

$$= cn \left[\frac{16^{\log_2(n)} - 1}{16 - 1} \right] = \frac{cn}{15} \left[n^{\log_2(16)} - 1 \right]$$

$$= \frac{cn}{15} \left[n^4 - 1 \right]$$

Thus, $T(n) \in \Theta(n^5)$

(b)

$$T(n) = \sum_{\alpha=0}^{n/4} 6n \cdot \left(\frac{3^\alpha}{4^\alpha}\right)^2$$

$$T(n) = 6n \sum_{\alpha=0}^{n/4} \left(9^\alpha / 16^\alpha\right)$$

$$= 6n \sum_{\alpha=0}^{n/4} \left(9/16\right)^\alpha$$

<u>UB</u>: $T(n) \leq 6n \sum_{\alpha=0}^{\infty} (9/16)^\alpha = 6n \cdot \left[\frac{1}{1 - 9/16}\right] = 6n \cdot \frac{16}{7}$

<u>LB</u>: $T(n) \geq 6n \, (9/16)^0 = 6n.$

Thus, $T(n) \in \Theta(n)$.

Problem 3.3. Determine the asymptotic complexity of the following function.

Hint: Bounding will help to make this problem **much** less terrible.

```
function Func(A[],n)
    x = 0
    for i = 3n^2 + 2n to 4n^3 + 2n + 3 do
        for j = 2i + 6 to 4i^2 + 2i + 5 do
            for k = ij to i^2 j do
                x = x + i - jk
            end for
        end for
    end for
    return x
end function
```

This problem is symbolically awful. I am sorry. You will never be asked to do this again.

I **strongly** recommend not computing actual values (i.e. leave constants as 11^5 instead of 161051)

$$T(n) = \sum_{i=3n^2+2n}^{4n^3+2n+3} \sum_{j=2i+6}^{4i^2+2i+5} \sum_{k=ij}^{i^2 j} c$$

$$= \sum_{i=3n^2+2n}^{4n^3+2n+3} \sum_{j=2i+6}^{4i^2+2i+5} c(i^2 j - ij + 1)$$

UB:

$$T(n) \le \sum_{i=1}^{9n^3} \sum_{j=1}^{11i^2} (i^2 j)$$

$$T(n) \le \sum_{i=1}^{9n^3} \sum_{j=1}^{11i^2} i^2(11i^2)$$

$$T(n) \le \sum_{i=1}^{9n^3} 11i^2 \cdot 11i^4$$

$$T(n) \le \sum_{i=1}^{9n^3} 11^2 (9n^3)^6$$

$$T(n) \le 9n^3 \cdot 11^2 \cdot 9^6 n^{18}$$

LB:

$$T(n) \ge \sum_{i=n^3}^{4n^3} \sum_{j=i^2}^{4i^2} (i^2 j - ij)$$

$$T(n) \ge \sum_{i=n^3}^{4n^3} \sum_{j=i^2}^{4i^2} i^2(i^2 - i)$$

$$i^4 - i^3 \ge i^4 - \frac{i^4}{2}$$

$$T(n) \ge \sum_{i=n^3}^{4n^3} (4i^2 - i^2)(i^4 - i^3)$$

$$T(n) \ge \sum_{i=n^3}^{4n^3} 3i^6/2$$

$$T(n) \ge (4n^3 - n^3) \cdot \frac{3(n^3)^6}{2}$$

Thus,

$$T(n) \in \Theta(n^{21}).$$